Density estimation on small data sets: Supplemental material

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SM.1 The posterior action $S_\ell[\phi]$

A derivation for Eq. 2 has already been reported in Ref. [1]. The derivation presented here, however, is more straightforward. The action in Eq. 2 is given by

$$S_\ell[\phi] = S_\ell^0[\phi] + S_{\text{data}}[\phi], \tag{S1}$$

where $S_\ell^0[\phi]$ is the "prior action", corresponding to a Bayesian prior $p(Q|\ell) \propto \exp(-S_\ell^0[\phi])$, while $S_{\text{data}}[\phi]$, the "likelihood action", is related to likelihood via $p(\text{data}|Q) \propto \exp(-S_{\text{data}}[\phi])$. DEFT uses a prior action of the form

$$S_\ell^0[\phi] = \int \frac{dx}{L} \frac{\ell^{2\alpha}}{2} (\partial^\alpha \phi)^2. \tag{S2}$$

The parameter α reflects a fundamental choice in how one defines "smoothness", and ℓ is a length scale below which fluctuations in ϕ are strongly damped. The derivation of $S_{\rm data}[\phi]$ is as follows. Suppose we are given N data points drawn from a probability distribution $Q_{\rm true}(x)$ that is confined to the interval $[x_{\rm min}, x_{\rm max}]$. Label these data in order of increasing value as $x_1, x_2, \ldots, x_N$. Next, imagine these data as being produced by a stochastic process in time, with x being the time variable and $r(x)$ being the instantaneous emission rate. The likelihood of the data is then given by

$$\begin{aligned}
(dx)^N p(\text{data}|r) &= \left[e^{-\int_{x_{\rm min}}^{x_1} dx\, r(x)}\right] \cdot [dx\, r(x_1)] \cdot \left[e^{-\int_{x_1}^{x_2} dx\, r(x)}\right] \cdot [dx\, r(x_2)] \;\cdots\; [dx\, r(x_N)] \cdot \left[e^{-\int_{x_N}^{x_{\rm max}} dx\, r(x)}\right] \\
&= (dx)^N \;\exp\left\{-\int_{x_{\rm min}}^{x_{\rm max}} dx\, r(x)\right\} \prod_{i=1}^{N} r(x_i) \\
&= (dx)^N \;\exp\left\{-\int dx\, r(x) + \sum_{i=1}^{N} \log\, r(x_i)\right\} \\
&= (dx)^N \;\exp\left\{-\int dx\, [r(x) - NR(x) \log\, r(x)]\right\}, \qquad \text{(S3)}
\end{aligned}$$

where $\int dx$ indicates integration over the entire x domain and $R(x) = N^{-1} \sum_{i=1}^{N} \delta(x - x_i)$ is the raw data density referred to in the main text. Next, we parametrize the emission rate $r(x)$ using the field $\phi(x)$ via

$$r(x) = \frac{N}{L} e^{-\phi(x)}. \qquad \text{(S4)}$$

The probability density corresponding to this rate is

$$Q(x) = \frac{r(x)}{\int dx'\, r(x')} = \frac{e^{-\phi(x)}}{\int dx'\, e^{-\phi(x')}}, \qquad \text{(S5)}$$

and so our definition of ϕ here is consistent with the definition of ϕ in the main text. We therefore see that the likelihood density in Eq. S3 is given by $p(\text{data}|\phi) \propto \exp(-S_{\rm data}[\phi])$ where the corresponding action (after dropping the constant term $N \log(L/N)$) is,

$$S_{\rm data}[\phi] = \int \frac{dx}{L} \left[NLR(x)\phi(x) + Ne^{-\phi(x)}\right]. \qquad \text{(S6)}$$

Plugging Eq. S2 and Eq. S6 into Eq. S1 gives Eq. 2 of the main text. Note the origin of the two terms in the integrand in Eq. S6: The term linear in ϕ comes from the exact locations of the N data points, whereas the nonlinear term (which leads to such interesting behavior) comes from regions of the x domain in which no data is observed.

We briefly discuss a subtle issue with the above derivation. The probability distribution $Q(x)$ is invariant under additive shifts in the underlying field, i.e., $\phi(x) \to \phi(x) + c$ for any constant c. By contrast, the likelihood action $S_{\rm data}[\phi]$ is not invariant under such transformations. This difference is due to Eq. S4 which, by specifying how the emission rate $r(x)$ relates to $\phi(x)$, introduces an additional assumption about how ϕ should be constrained by data. But although this additional assumption alters $p(\phi|\text{data})$, it does not alter $p(Q|\text{data})$. The more involved derivation of $S_\ell[\phi]$ provided in Ref. [1] demonstrates this fact explicitly.

SM.2 The MAP field ϕ_ℓ

To solve for ϕ_ℓ, the maximum *a posteriori* (MAP) field at length scale ℓ, we set $\delta S_\ell / \delta\phi = 0$. The resulting equation of motion is

$$\ell^{2\alpha} \Delta^\alpha \phi_\ell + NLR - Ne^{-\phi_\ell} = 0. \qquad \text{(S7)}$$

The operator Δ^α that appears here is the "bilateral Laplacian", which is described in Ref. [1]. Briefly, Δ^α is defined by the requirement that

$$\int dx\; \varphi \Delta^\alpha \phi = \int dx\; (\partial^\alpha \varphi)(\partial^\alpha \phi), \qquad \text{(S8)}$$

for any two fields φ and ϕ. This bilateral Laplacian is identical to the standard α-order Laplacian $(-1)^\alpha \partial^{2\alpha}$ in the interior of the x interval, but differs at the boundaries. Specifically, the standard α-order Laplacian requires the additional specification of 2α boundary conditions in order to be self-adjoint. By contrast, the bilateral Laplacian is self-adjoint without the specification of any boundary conditions. The equation of motion, Eq. S7, thus has a unique solution without the need to assume any boundary conditions on ϕ. See Ref. [1] for more information.

By integrating Eq. S7 we find that $\int dx\, e^{-\phi_\ell(x)} = L$, due to $\int dx\, R(x) = 1$ and $\int dx \Delta^\alpha \phi_\ell = \int dx (\partial^\alpha 1)(\partial^\alpha \phi_\ell) = 0$. The MAP density Q_ℓ thus has a simple form:

$$Q_\ell(x) = \frac{e^{-\phi_\ell(x)}}{L}. \tag{S9}$$

Similarly, multiplying Eq. S7 on the left by x^k for $k = 1, \ldots, \alpha - 1$ and integrating reveals that

$$\left\langle x^k \right\rangle_{Q_\ell} = \left\langle x^k \right\rangle_R, \tag{S10}$$

i.e., the first $\alpha - 1$ moments of Q_ℓ exactly match those of the data.

As described in Ref. [1], DEFT computes the MAP field ϕ_ℓ for a set of length scales $\{\ell_0, \ell_1, \ell_2, \ldots, \ell_K\}$, ranging from $\ell_0 = 0$ to $\ell_K = \infty$. These length scales are chosen so that neighboring MAP densities, Q_{ℓ_k} and $Q_{\ell_{k+1}}$, are approximately equally spaced along this "MAP curve", as quantified by the geodesic distance $D_{\text{geo}}(Q_{\ell_k}, Q_{\ell_{k+1}})$. We note that Q_0 is in fact the data histogram R, while Q_∞ is in fact the maximum entropy distribution consistent with the moment constraints in Eq. S10. See Ref. [1] for details.

SM.3 The evidence $p(\text{data}|\ell)$

The DEFT algorithm computes the MAP field at length scales spanning $\ell = 0$ to $\ell = \infty$. The optimal length scale ℓ^* is then computed by maximizing the Bayesian evidence $p(\text{data}|\ell)$. The key quantity needed for this procedure is the "evidence ratio", which is given by

$$E(\ell) = \frac{p(\text{data}|\ell)}{p(\text{data}|\infty)}. \tag{S11}$$

It can be shown that $E(\ell) = (Z_\ell / Z_\ell^0)/(Z_\infty / Z_\infty^0)$, where

$$Z_\ell = \int \mathcal{D}\phi\; e^{-S_\ell[\phi]} \quad \text{and} \quad Z_\ell^0 = \int \mathcal{D}\phi\; e^{-S_\ell^0[\phi]} \tag{S12}$$

respectively denote the posterior partition function and the prior partition function. The prior partition function Z_ℓ^0 can be computed analytically, although it has a divergence that must be regularized. By contrast, the posterior partition function Z_ℓ can only be analytically computed in the Laplace approximation. We therefore instead use the quantity

$$Z_\ell^{\text{Lap}} = \int \mathcal{D}\phi\; e^{-S_\ell^{\text{Lap}}[\phi]}, \tag{S13}$$

where $S_\ell^{\text{Lap}}[\phi]$ is the Laplace approximation of $S_\ell[\phi]$. The resulting evidence ratio in this approximation is found to be

$$E(\ell) = e^{S_\infty[\phi_\infty] - S_\ell[\phi_\ell]} \sqrt{\frac{\det_{\text{ker}}[e^{-\phi_\infty}] \det_{\text{row}}[L^{2\alpha}\Delta^\alpha]}{\eta^{-\alpha} \det[L^{2\alpha}\Delta^\alpha + \eta e^{-\phi_\ell}]}}, \tag{S14}$$

where $\eta = N(L/\ell)^{2\alpha}$, and "ker" and "row" respectively denote the kernel and row space of the bilateral Laplacian Δ^α. See Ref. [1] for details.

It should be emphasized that, although the Laplace approximation can be grossly innacurate when sampling $Q \sim p(Q|\text{data})$, it does not strongly affect the evidence ratio $E(\ell)$. This is because $\log E(\ell)$ typically varies over many orders of magnitude, whereas $\log(Z_\ell / Z_\ell^{\text{Lap}})$ varies with ℓ far less dramatically. This is demonstrated in Fig. S2 below. Nevertheless, the SUFTware implementation of DEFT includes an option to correct for this approximation using importance sampling, as described in Eq. 9 the main text.

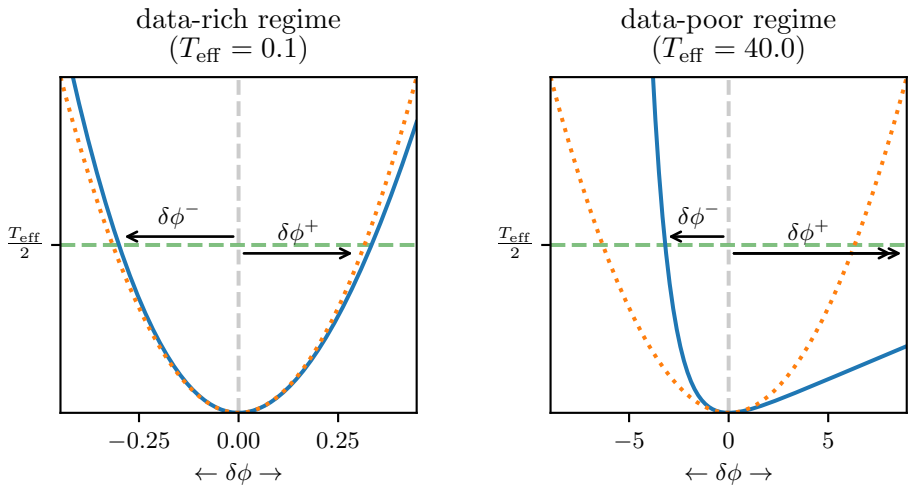

Figure S1: **Fluctuations in data-rich vs. data-poor regimes**. Solid blue lines indicate $f(\delta\phi)$ from Eq. S18. Dotted orange lines indicate the Laplace approximation $f_{\rm Lap}(\delta\phi) = \delta\phi^2/2$. Dashed green lines indicate the value of half the effective temperature ($T_{\rm eff}/2$) from Eq. S20. In the data-rich regime, $T_{\rm eff}/2 \ll 1$ (left panel), resulting in nearly symmetric $\delta\phi^\pm$ fluctuations. In the data-poor regime, $T_{\rm eff}/2 \gg 1$ (right panel), resulting in highly asymmetric fluctuations; in particular, the magnitude of $\delta\phi^-$ due to f is substantially less than would result from $f_{\rm Lap}$. Note also that the very large positive fluctuations $\delta\phi^+$ in the data-poor regime have little noticeable effect on Q_ℓ, since they just push Q_ℓ closer to zero.

SM.4 Sampling the posterior $p(\phi, \ell|\text{data})$

The posterior probability $p(\phi, \ell|\text{data})$ can be decomposed as

$$p(\phi, \ell|\text{data}) = p(\phi|\ell, \text{data})\ p(\ell|\text{data}). \tag{S15}$$

This forms the basis for our posterior sampling procedure. First, we sample plausible ℓs from $p(\ell|\text{data})$. Note that $p(\ell|\text{data}) \propto p(\text{data}|\ell)\ p(\ell)$ by Bayes's Theorem. Assuming $p(\ell)$ is uniform over the length of the MAP curve as quantified by geodesic distance (see Ref. [1]), $p(\ell|\text{data})$ becomes proportional to the evidence ratio $E(\ell)$. We thus sample values of ℓ from the set $\{\ell_0, \ell_1, \ldots, \ell_K\}$ used to trace the MAP curve, each ℓ_k being selected with probability proportional to $E(\ell_k)$. For each of these ℓ values, we then sample plausible ϕs from $p(\phi|\ell, \text{data})$. Here we employ importance sampling. Specifically, we can rewrite the distribution $p(\phi|\ell, \text{data})$ as follows

$$p(\phi|\ell, \text{data}) = \frac{e^{-S_\ell[\phi]}}{Z_\ell} = \frac{e^{-S_\ell^{\rm Lap}[\phi]}}{Z_\ell^{\rm Lap}} \frac{w_\ell[\phi]}{\langle w_\ell \rangle_{{\rm Lap}|\ell}} \propto p_{\rm Lap}(\phi|\ell, \text{data}) w_\ell[\phi], \tag{S16}$$

where we have made use of Eq. S36 (derived below). Therefore, we first sample ϕs from the Laplace-approximated distribution $p_{\rm Lap}(\phi|\text{data}, \ell)$, then correct for the non-Gaussian nature of the original distribution by resampling these ϕs using the importance weights $w_\ell[\phi]$.

SM.5 Origin of wisps

To derive Eqs. 4 and 5, it suffices to note that

$$S_\ell[\phi_\ell + \delta\phi] \quad = \quad S_\ell[\phi_\ell] + O(\delta\phi^2), \tag{S17}$$

because the equation of motion, Eq. S7, causes all first-order terms in $\delta\phi$ to cancel. Next, we express $V(\delta\phi) = NLQ_\ell f(\delta\phi)$ where

$$f(\delta\phi) = e^{-\delta\phi} - 1 + \delta\phi \tag{S18}$$

is just $e^{-\delta\phi}$ with the 0th and 1st order terms subtracted out. This function is plotted in Fig. S1. The key to deriving the magnitude of fluctuations $\delta\phi$ in different regimes is the relationship $\langle V(\delta\phi)\rangle = \frac{n_{\text{eff}}}{2}$, where n_{eff} is the effective number of degrees of freedom. We reexpress this as

$$\langle f(\delta\phi)\rangle = \frac{T_{\text{eff}}}{2}, \tag{S19}$$

where

$$T_{\text{eff}} = \frac{n_{\text{eff}}}{NLQ_\ell} \tag{S20}$$

is an effective temperature.

In the data-rich regime, $T_{\text{eff}}/2 \ll 1$. Therefore, $f(\delta\phi) \ll 1$ for typical fluctuations $\delta\phi$. As illustrated in Fig. S1 (left panel), the Laplace approximation works well in this regime. Setting

$$f(\delta\phi) \approx f_{\text{Lap}}(\delta\phi) = \frac{\delta\phi^2}{2} \sim \frac{T_{\text{eff}}}{2} \tag{S21}$$

and solving for $\delta\phi$ gives Eq. 6.

In the data-poor regime, $T_{\text{eff}}/2 \gg 1$. As illustrated in Fig. S1 (right panel), f is highly asymmetric in this regime and so the positive and negative fluctuations, $\delta\phi^+$ and $\delta\phi^-$, need to be treated separately. Specifically,

$$f(\delta\phi^+) \approx \delta\phi^+ \sim \frac{T_{\text{eff}}}{2}, \quad \text{whereas} \quad f(\delta\phi^-) \approx e^{-\delta\phi^-} \sim \frac{T_{\text{eff}}}{2}. \tag{S22}$$

Solving the latter condition for $\delta\phi^-$ gives Eq. 7. Note in Fig. S1 (right panel) how the Laplace approximation greatly overestimates the magnitude of negative fluctuations $\delta\phi^-$ in the data-poor regime.

SM.6 Computing $\log(Z_\ell/Z_\ell^{\text{Lap}})$ using Feynman diagrams

Here we show how Feynman diagrams can be used to compute $\log(Z_\ell/Z_\ell^{\text{Lap}})$, thereby obtaining corrections to the Laplace approximation. Our exposition closely follows that sketched by Zinn-Justin [2]. However, because Feynman diagrams are rarely used in the context of statistical inference, we have chosen to make these calculations explicit.

Upon discretization of the x interval using G grid points, the action in Eq. 2 becomes

$$S_\ell[\phi] = \frac{\ell^{2\alpha}}{2G}\sum_{ij}\Delta^\alpha_{ij}\phi_i\phi_j + \frac{NL}{G}\sum_i R_i\phi_i + \frac{N}{G}\sum_i e^{-\phi_i}, \tag{S23}$$

where $i, j = 1, 2, \ldots, G$. In what follows we represent fluctuations in ϕ about the MAP field (here denoted ϕ^ℓ) using the rescaled fluctuation $x = \sqrt{N}(\phi - \phi^\ell)$. The action can then be expanded in the following way:

$$S_\ell[\phi] = S_\ell^{\text{Lap}}[\phi] + \frac{1}{3!}\sum_{ijk}\frac{B_{ijk}}{\sqrt{N}}x_ix_jx_k + \frac{1}{4!}\sum_{ijkl}\frac{C_{ijkl}}{N}x_ix_jx_kx_l + \cdots, \tag{S24}$$

where the Laplace action is

$$S_\ell^{\text{Lap}}[\phi] = S_\ell[\phi^\ell] + \frac{1}{2}\sum_{ij}A_{ij}x_ix_j, \tag{S25}$$

and

$$A_{ij} = \frac{1}{N}\left.\frac{\partial^2 S_\ell}{\partial\phi_i\partial\phi_j}\right|_{\phi^\ell} = \frac{\ell^{2\alpha}}{NG}\Delta^\alpha_{ij} + \frac{1}{G}e^{-\phi_i^\ell}\delta_{ij}, \tag{S26}$$

$$B_{ijk} = \frac{1}{N}\left.\frac{\partial^3 S_\ell}{\partial\phi_i\partial\phi_j\partial\phi_k}\right|_{\phi^\ell} = -\frac{1}{G}e^{-\phi_i^\ell}\delta_{ijk}, \tag{S27}$$

$$C_{ijkl} = \frac{1}{N}\left.\frac{\partial^4 S_\ell}{\partial\phi_i\partial\phi_j\partial\phi_k\partial\phi_l}\right|_{\phi^\ell} = \frac{1}{G}e^{-\phi_i^\ell}\delta_{ijkl}. \tag{S28}$$

The quantity $\log(Z_\ell/Z_\ell^{\rm Lap})$ is conveniently given by the sum of connected vacuum diagrams. At $O(N^{-1})$, the relevant diagrams contain only 3rd-order and 4th-order vertices. From the expansion in Eq. S24 we see that the values corresponding to these vertices are given by $-B_{ijk}/\sqrt{N}$ and $-C_{ijkl}/N$, respectively. We also need the propagator matrix P, which is given by the inverse of the Hessian A, i.e., $P_{ij} = (A^{-1})_{ij}$. We thus obtain

$$\log \frac{Z_\ell}{Z_\ell^{\rm Lap}} =$$ $+$ $+$ $+ O(N^{-2}),$ (S29)

where the contribution from each diagram is

$$= \frac{1}{8}\sum_{ijkl}\left(-\frac{C_{ijkl}}{N}\right)P_{ij}P_{kl} = -\sum_i \frac{e^{-\phi_i^\ell}}{8NG}\left(P_{ii}\right)^2, \tag{S30}$$

$$= \frac{1}{8}\sum_{ijk}\sum_{lmn}\left(-\frac{B_{ijk}}{\sqrt{N}}\right)\left(-\frac{B_{lmn}}{\sqrt{N}}\right)P_{ij}P_{kl}P_{mn} = \sum_i\sum_l \frac{e^{-\phi_i^\ell-\phi_l^\ell}}{8NG^2}P_{ii}P_{il}P_{ll}, \tag{S31}$$

$$= \frac{1}{12}\sum_{ijk}\sum_{lmn}\left(-\frac{B_{ijk}}{\sqrt{N}}\right)\left(-\frac{B_{lmn}}{\sqrt{N}}\right)P_{il}P_{jm}P_{kn} = \sum_i\sum_l \frac{e^{-\phi_i^\ell-\phi_l^\ell}}{12NG^2}\left(P_{il}\right)^3. \tag{S32}$$

SM.7 Computing $\log(Z_\ell/Z_\ell^{\rm Lap})$ using importance sampling

Alternatively, the correction $\log(Z_\ell/Z_\ell^{\rm Lap})$ can be computed using importance sampling involving the weights w_ℓ in Eq. 3 of the main text. To see how, we express the partition function Z_ℓ as an average over the Laplace ensemble:

$$\begin{aligned} Z_\ell &= \int \mathcal{D}\phi\; e^{-S_\ell[\phi]} && \text{(S33)}\\ &= Z_\ell^{\rm Lap}\int \mathcal{D}\phi\; \frac{e^{-S_\ell^{\rm Lap}[\phi]}}{Z_\ell^{\rm Lap}}\; e^{S_\ell^{\rm Lap}[\phi]-S_\ell[\phi]} && \text{(S34)}\\ &= Z_\ell^{\rm Lap}\int \mathcal{D}\phi\; p_{\rm Lap}(\phi|{\rm data},\ell)\; w_\ell[\phi] && \text{(S35)}\\ &= Z_\ell^{\rm Lap}\;\langle w_\ell\rangle_{{\rm Lap}|\ell}, && \text{(S36)} \end{aligned}$$

where $\langle\cdot\rangle_{{\rm Lap}|\ell}$ denotes the mean taken with respect to the Laplace posterior $p_{\rm Lap}(\phi|{\rm data},\ell)$, and w_ℓ denotes the importance sampling weights in Eq. 3. The quantity $\log(Z_\ell/Z_\ell^{\rm Lap})$ can thus be computed using Eq. 9 of the main text.

SM.8 Feynman diagrams vs. importance sampling

Perhaps disappointingly, Feynman diagrams generally do not work well in situations where wisps appear. This is because the posterior action in such cases is strongly coupled. To see this, consider an expansion of the potential V in Eq. 5 of the main text to m'th order in $\delta\phi$:

$$V_m(\delta\phi) = NLQ_\ell\sum_{n=2}^{m}\frac{(-\delta\phi)^n}{n!}. \tag{S37}$$

To produce accurate results, the potential V_m must include enough terms to sufficiently approximate V when evaluated at $\delta\phi = -\delta\phi^-_{\rm poor} = -\phi^* + \log(N/n_{\rm eff})$. This would require $m_{\rm min} = \phi^* - \log(N/n_{\rm eff})$ terms at the very least, since not until here do the (all positive) terms in this power series begin to decrease. Thus, the number of terms that would be needed cannot be fixed *a priori*, but rather must increase with ϕ^*. This presents a major problem for Feynman-diagram-based expansions. Any diagram influenced by the $m_{\rm min}$'th term in Eq. S37 must contain an $m_{\rm min}$'th order vertex. But $m_{\rm min}$ can be quite large: For ϕ^* in Fig. 1, one

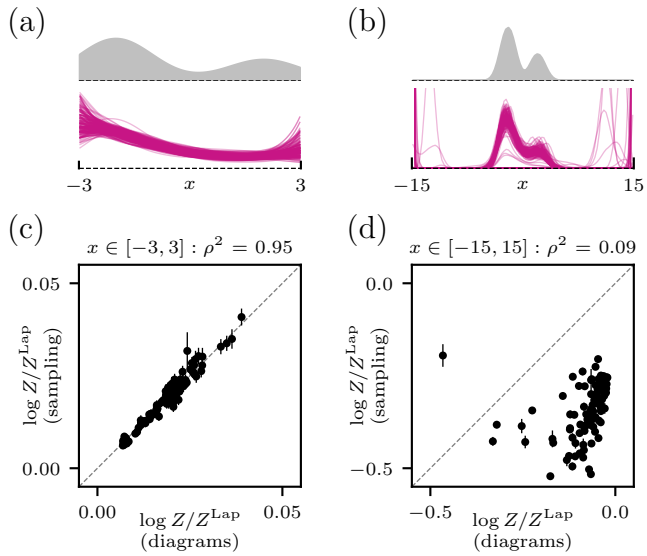

Figure S2: **Wisps appear only when the posterior action is strongly coupled**. The accuracy of Feynman diagrams was assessed using data drawn from the Q_{true} density in Fig. 1 confined to the intervals $[-3, 3]$ (a,c) or $[-15, 15]$ (b,d). (a,b) Q_{true} (gray) is shown along with 100 distributions Q (magenta) sampled at fixed $\ell = \ell^*$ from the Laplace-approximated posterior inferred from a data set of size $N = 100$. Wisps are observed in (b) but not in (a). (c,d) Values for $\log(Z_\ell/Z_\ell^{\text{Lap}})$ computed for 100 different data sets, generated as above, using either Feynman diagrams (Eq. 8) or importance weights (Eq. 9). These two quantities agree well in (c) but poorly in (d). Squared Pearson correlations, ρ^2, are shown in the titles of (c,d).

finds $m_{\text{min}} > 100$ near the boundaries of the x interval. Evaluating Feynman diagrams up to such high order is not feasible.

This expectation is confirmed in Fig. S2, which compares the two ways of computing $\log(Z_\ell/Z_\ell^{\text{Lap}})$ for two different choices of Q_{true}. The Feynman diagram approximation works well when Q_{true} fills the entire x interval, indicating that the action $S_\ell[\phi]$ is nearly quadratic and the corrections to the Laplace approximation are small. However, when Q_{true} vanishes in large regions of the x domain, the Feynman diagram approximation is very bad. In this case, the action $S_\ell[\phi]$ is strongly coupled and a fundamentally nonperturbative approach is required to compute the corrections.

Although the nonquadratic nature of the posterior action can lead to a partition function Z_ℓ differing from its Laplace-approximated value Z_ℓ^{Lap} by a large amount, we find that the Laplace approximation generally works well nevertheless for identifying the optimal length scale. This is because $\log Z_\ell^{\text{Lap}}$ typically varies by multiple orders of magnitude across different values of ℓ, thereby swamping potential inaccuracies in the $Z_\ell \approx Z_\ell^{\text{Lap}}$ assumption.

SM.9 Other density estimation methods

Here we describe the Kernel density estimation (KDE) and Dirichlet process mixture modeling (DPMM) algorithms used for the computations shown in Fig. 2 and Fig. S3.

SM.9.1 Kernel density estimation (KDE)

KDE is arguably the most common approach to density estimation in one dimension. Given data $\{x_i\}_{i=1}^N$, the KDE density estimate is given by

$$Q^*(x) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{w}K\left(\frac{x - x_i}{w}\right), \tag{S38}$$

where $K(z)$ is the kernel function and w is the "bandwidth". We used a Gaussian kernel,

$$K(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \tag{S39}$$

and chose the bandwidth w using cross-validation. Specifically, we considered 100 candidate bandwidths geometrically distributed between $w_{\min}$ (the minimum spacing between data points) and $w_{\max}$ (10 times the span of the data). We then chose the bandwidth w that maximized the jackknifed log likelihood

$$\mathcal{L}(w) = \sum_{i=1}^{N} \log Q^*_{-i}(x_i), \tag{S40}$$

where the subscript on Q^*_{-i} indicates the density Q^* computed as in Eq. S38 but using a data set missing the datum x_i.

KDE does not provide an explicit posterior on Q. Therefore, to compute p-values for Fig. 2 and Fig. S3, we approximated posterior samples $Q \sim p(Q|\text{data})$ by applying KDE to bootstrap-resampled data sets.

SM.9.2 Dirichlet process mixture modeling (DPMM)

DPMM is arguably the most popular nonparametric Bayesian method for estimating probability densities. DPMMs have a hierarchical structure, in the sense that each data point is assumed to be drawn from one of a number of "clusters", with each cluster having a probability density defined by a kernel of pre-specified functional form.

In the computations for Fig. 2 and Fig. S3, we adopted the finite DPMM described in Refs. [3, 4]. Densities were assumed to be of the form

$$Q(x) = \sum_{h=1}^{H} w_h K_{m_h}(x), \tag{S41}$$

where H is the number of clusters, w_h is the probability of cluster h, and m_h is the set of parameters defining the density of cluster h. $K_m(z)$ was assumed to be a Gaussian density specified by $m = (\mu, \sigma^2)$, i.e., a mean and a variance. A normal-inverse-gamma distribution was used as the prior on m:

$$p(\mu, \sigma^2) = \mathcal{N}(\mu|\hat{\mu}, \hat{\kappa}\sigma^2)\ \Gamma^{-1}(\sigma^2|\hat{\alpha}, \hat{\beta}), \tag{S42}$$

where $\hat{\kappa} = 1$, $\hat{\alpha} = 1$, $\hat{\beta} = \hat{\sigma}^2$,

$$\hat{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \hat{\mu})^2, \quad \text{and} \quad \hat{\mu} = \frac{1}{N} \sum_{i=1}^{N} x_i. \tag{S43}$$

The number of clusters was fixed at $H = 10$. For each data set, we used Gibbs sampling to obtain an ensemble of plausible densities representing $p(Q|\text{data})$. The optimal estimate Q^* was then defined as the mean density in this ensemble. Following Ref. [4], our Gibbs sampling algorithm worked as follows. For each cluster $h = 1, 2, \ldots, H$, we chose an initial weight $w_h = 1/H$ and a set of kernel parameters m_h selected according to the prior distribution $p(\mu, \sigma^2)$ in Eq. S42. The sampler was then run by iterating the following steps:

1. Data were redistributed across clusters. Specifically, each data point x_i was allocated to cluster h with probability

$$p(h|x_i) = \frac{w_h K_{m_h}(x_i)}{\sum_{h'=1}^{H} w_{h'} K_{m_{h'}}(x_i)}. \tag{S44}$$

2. The mean and variance of each cluster were updated using

$$m_h \sim \mathcal{N}(\mu_h|\hat{\mu}_h, \hat{\kappa}_h \sigma_h^2)\ \Gamma^{-1}(\sigma_h^2|\hat{\alpha}_h, \hat{\beta}_h), \tag{S45}$$

where

$$\hat{\mu}_h = \hat{\kappa}_h \left(\frac{\hat{\mu}}{\hat{\kappa}} + n_h \langle x \rangle_h \right), \tag{S46}$$

$$\hat{\kappa}_h = \frac{\hat{\kappa}}{1 + n_h \hat{\kappa}}, \tag{S47}$$

$$\hat{\alpha}_h = \hat{\alpha} + \frac{n_h}{2}, \tag{S48}$$

$$\hat{\beta}_h = \hat{\beta} + \frac{1}{2} \left(\sum_{i \epsilon h} (x_i - \langle x \rangle_h)^2 + \frac{n_h}{1 + n_h \hat{\kappa}} (\langle x \rangle_h - \hat{\mu})^2 \right). \tag{S49}$$

Here, $\langle x \rangle_h$ represents the mean value of data points belonging to cluster h, and n_h is the number of data points in this cluster.

3. The cluster weights were updated by sampling

$$w_1, \ldots, w_H \sim \text{Dirichlet}(1 + n_1, \ldots, 1 + n_H). \tag{S50}$$

SM.10 Computational complexity

An explicit expression for the algorithmic complexity of DEFT is not very helpful for understanding runtime performance. This is because DEFT involves multiple steps computed in series, the runtimes of which are governed by different parameters. In practice, we have found DEFT to be primarily limited by the number of grid points G. This is because a computation of the evidence ratio $E(\ell)$, as well as posterior sampling, requires a spectral decomposition of the $G \times G$ Hessian matrix at each length scale ℓ along the MAP curve. We note, however, that DEFT computations with $G = 100$ are generally quite fast (i.e., ~ 0.25 seconds on a standard laptop computer). Although DEFT requires histogramming the data, which is $\mathcal{O}(N)$, this is rarely the bottleneck in practice. In fact, we have found that the speed of DEFT often *increases* with N, since this leads to a shorter MAP curve, thus requiring fewer discrete length scales ℓ to be examined.

In our computations for Fig. 2 and Fig. S3, DEFT was often faster than our KDE and DPMM implementations. The use of jackknife cross-validation greatly slows down KDE in a manner that increases linearly with N. DPMM, on the other hand, is greatly slowed by its reliance on Gibbs sampling, which is necessitated by the nonconvexity of the parameter posterior. In fact, Gibbs sampling is needed not just to generate a posterior sample, but also to estimate Q^* (via a posterior mean). We note that the accuracy of KDE and DPMM is also very sensitive to the choice of kernel, especially when data is clustered near the x interval boundaries.

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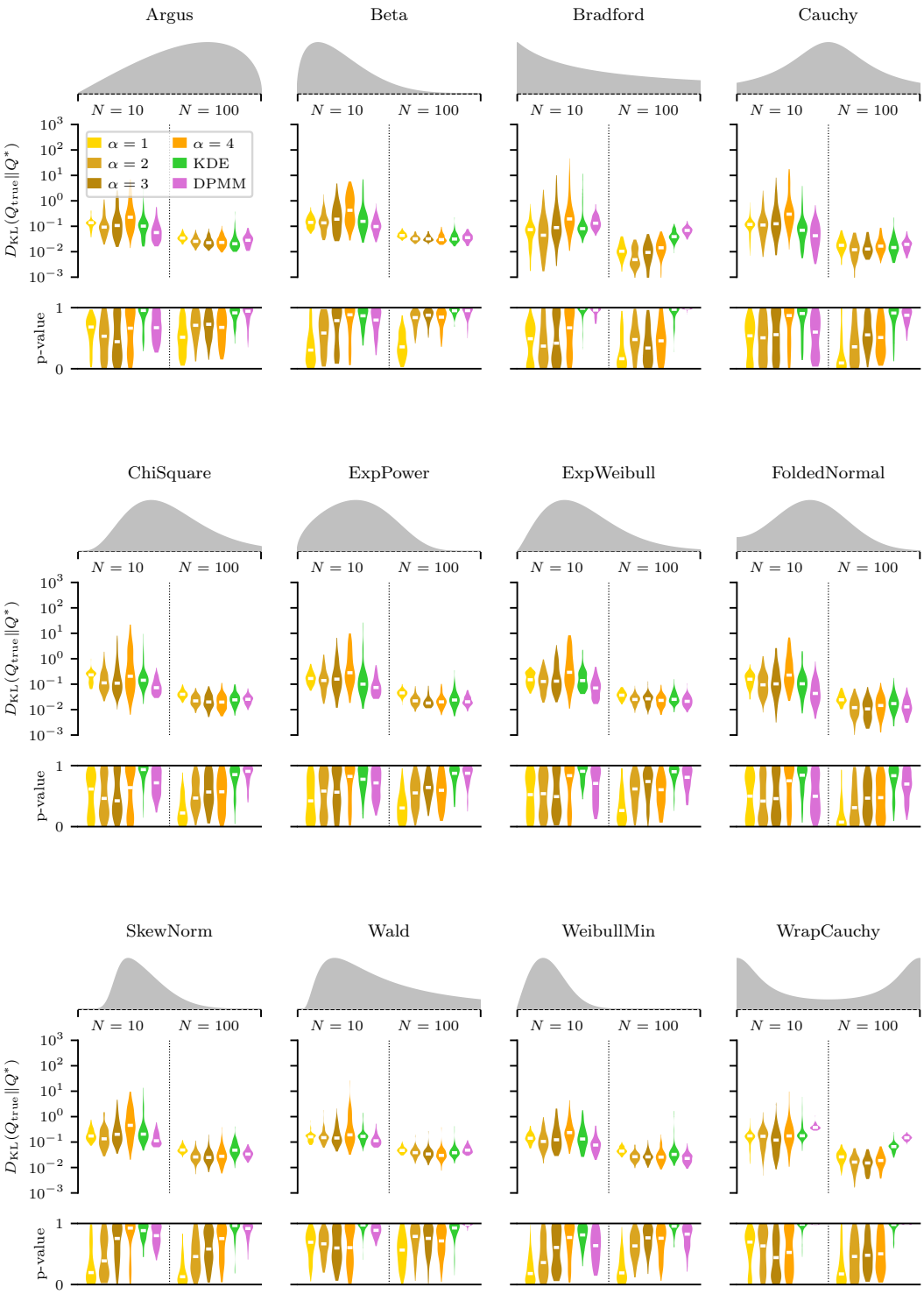

Figure S3: **Extension of Fig. 2 to other distributions**. The same analysis as in Fig. 2 was performed for twelve additional Q_{true} distributions, which were selected from the built-in distributions in the `scipy.stats` Python library.

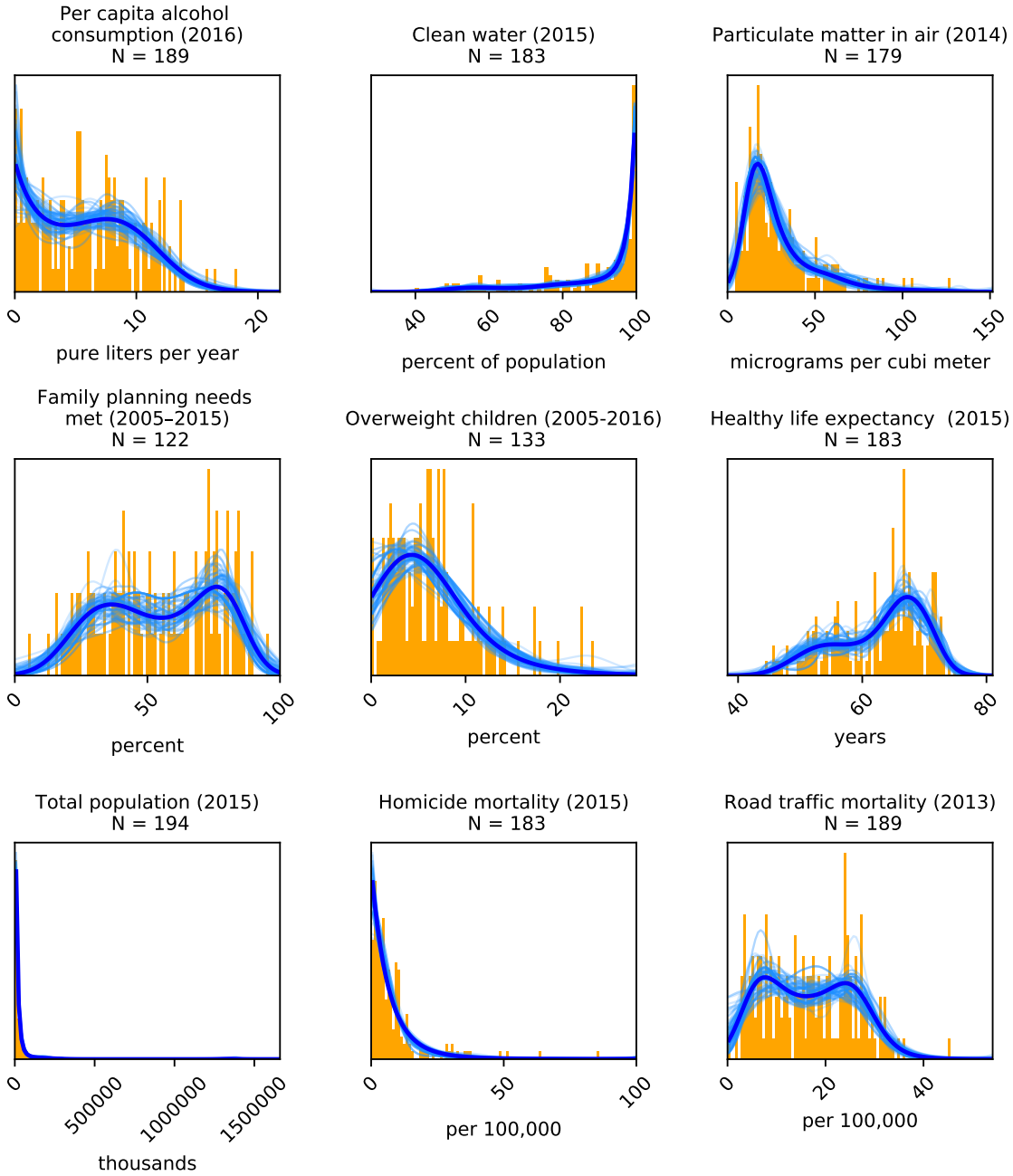

Figure S4: **Demonstration of DEFT on data from the World Health Organization (WHO)**. Densities were estimated for 9 different global health indicators reported by the WHO in [5]. Each datum corresponds to a different country; N varies between panels because of missing data in [5]. Orange shows a histogram of each global health indicator computed using $G = 100$ grid points. The best DEFT estimate Q^* is shown in dark blue, while 100 posterior-sampled densities $Q \sim p(Q|\text{data})$ are shown in light blue. As in Fig. 3, default DEFT parameters were used for all of these data sets.